

CO3 AT3
Scenario Based

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CSA1403

Compiler Design

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1)

2) syntax tree Analysis for $a * b - c + 8$.

$$E \rightarrow E + T$$

$$E \rightarrow E - T$$

$$E \rightarrow T$$

$$T \rightarrow T_1 * F$$

$$T \rightarrow F$$

$$F \rightarrow id$$

$$F \rightarrow digit$$

assume $a = 2, b = 3, c = 4, dis = 8$

$$F_1.val = id_a.val = 2$$

$$T_1.val = F_1.val = 2$$

$$F_2.val = id_b.val = 3$$

$$T_2.val = T_1.val * F_2.val = 6$$

$$E_1.val = T_2.val = 6$$

$$F_3.val = id_c.val = 4$$

$$T_3.val = F_3.val = 4$$

$$E_2.val = E_1.val - T_3.val = 2$$

$$F_4.val = digit.lexval = 8$$

$$T_4.val = F_4.val = 8$$

$$E_3.val = E_2.val + T_4.val = 10$$

3)

Rule 1:

=

$$P_i = A_i + 2$$

$$Q_i = P_i + A_i$$

$$P_5 = P_5 + Q_5$$

* P_i depends only on $A_i \rightarrow$ valid

$Q_i \rightarrow$ depends on P_i & A_i

$A_8 \rightarrow$ synthesized attribute depend on children.
 $\rightarrow$ valid

Rule 2:

$$A \rightarrow XV$$

$$X_i = A_i + Y_5$$

$$Y_i = X_5 + A_i$$

$\Rightarrow$ To compute $X_i \rightarrow$ rule attempts to use Y_5

$\Rightarrow$ + attributed definitions strictly forbid symbol depend on attribute of symbol.

conclude :

1st Rule satisfy requirements of L attribute

2nd rule $\rightarrow$ violates requirements of L-attributes.

1) s- attributed SDD :

$S \rightarrow EN$

$E \rightarrow E, + T$

$E \rightarrow E, - T$

$E \rightarrow T$

$T \rightarrow T, * F$

$T \rightarrow T, / F$

$T \rightarrow F$

$F \rightarrow (E)$

$F \rightarrow \text{digit}$

$N \rightarrow ;$

LR parse trace $2+5*6$

Stack

Input

Action

\$

$2+5*6 ; \$$

shift 2

$\$2$

$+5*6 ; \$$

Reduce $F \rightarrow \text{digit}$

$\$F$

$+5*6 ; \$$

Reduce $T \rightarrow F$

$\$T$

$+5*6 ; \$$

Shift Reduce $E \rightarrow T$

$\$E$

$+5*6 ; \$$

Shift +

$\$E+$

$5*6 ; \$$

shift 5

$\$E+5$

$*6 ; \$$

Reduce $F \rightarrow \text{digit}$

$\$E+F$

$*6 ; \$$

Reduce $T \rightarrow F$

$\$E+T$

$*6 ; \$$

Shift *

$\$E+T*$

$6 ; \$$

shift 6

$\$E+T*6$

$; \$$

Reduce $F \rightarrow \text{digit}$

$\$E+T*F$

$; \$$

Reduce $T \rightarrow T * F$

$\$E+T$

$; \$$

Reduce $E \rightarrow E+T$

$\$E$

$; \$$

Shift ;

$\$EN$

$\$$

Reduce $S \rightarrow EN$

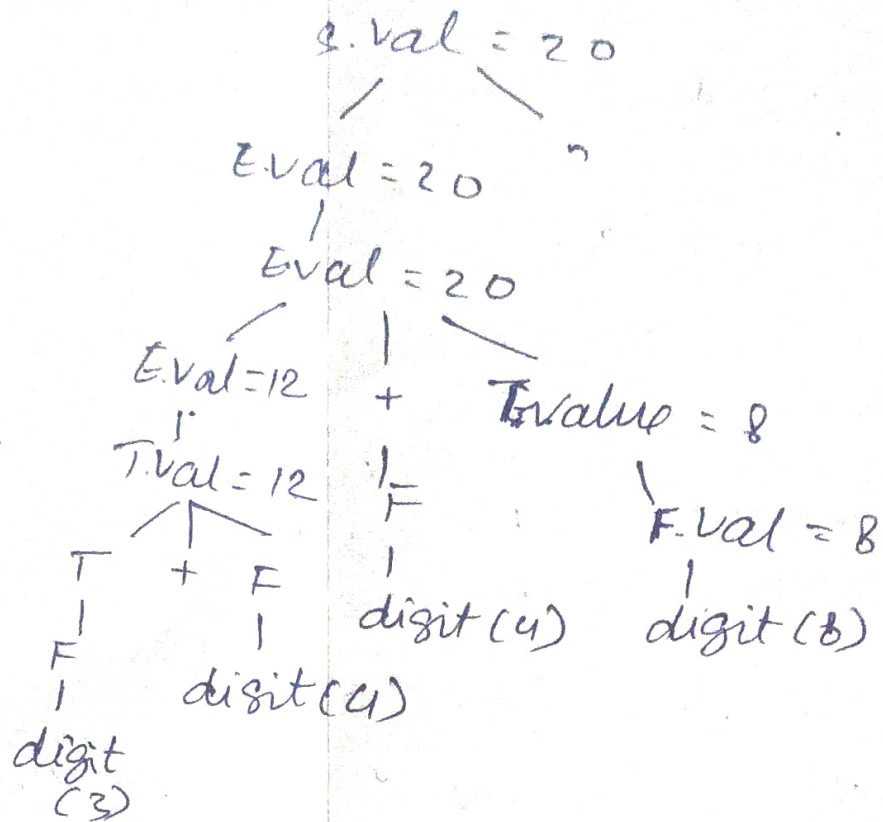
$\$S$

$\$$

Accept

5) SDD for Top down

$S \rightarrow E \mid$
 $E \rightarrow E + T$
 $E \rightarrow T$
 $T \rightarrow T * F$
 $T \rightarrow F$
 $F \rightarrow (E)$
 $F \rightarrow \text{digit}$



Final val = 20 for annotated parse tree for $3 * 4 + 8$.